

the outcome of the treated group. [Leavitt \(2020\)](#) proposes an empirical Bayes approach calibrated to pre-treatment differences in trends, and [Bilinski and Hatfield \(2020\)](#) and [Dette and Schumann \(2020\)](#) propose non-inferiority approaches based on pre-tests for the magnitude of the pre-treatment violations of parallel trends.

Our methods address several concerns related to current empirical practice in DiD and event-study designs. First, pre-trends tests may be underpowered against meaningful violations of parallel trends, potentially leading to severe undercoverage of conventional confidence intervals ([Freyaldenhoven et al., 2019](#); [Bilinski and Hatfield, 2020](#); [Kahn-Lang and Lang, 2020](#); [Roth, 2022](#)). Second, statistical distortions from pre-trends tests may further undermine the performance of conventional inference procedures ([Roth, 2022](#)). Third, parametric approaches to controlling for pre-existing trends may be sensitive to functional form assumptions ([Wolfers, 2006](#); [Lee and Solon, 2011](#)). We address these issues by providing tools for inference that do not rely on an exact parallel trends assumption, incorporate statistical uncertainty about the estimated event-study coefficients, and make clear the mapping between the researcher's assumptions about the potential differences in trends and the strength of their causal conclusions.

Our work complements a growing literature on the causal interpretation of event-study coefficients in two-way fixed effects models in the presence of staggered treatment timing and heterogeneous treatment effects ([Borusyak and Jaravel, 2016](#); [Callaway and Sant'Anna, 2020](#); [de Chaisemartin and D'Haultfœuille, 2020](#); [Sun and Abraham, 2020](#); [Athey and Imbens, 2021](#); [Goodman-Bacon, 2021](#)). A key finding is that regression coefficients from conventional approaches may not produce convex weighted averages of treatment effects even if parallel trends holds. Several alternative estimators have been proposed that consistently estimate interpretable causal estimands under a suitable parallel trends assumption. Our methodology can be used in conjunction with these alternative estimators to assess their sensitivity to violations of the corresponding parallel trends assumption; see [Section 2.1](#) for additional details.

More broadly, our work contributes to a larger econometric literature that uses partial identification to provide empirical researchers with tractable tools to conduct inference under assumptions that may be more credible in empirical practice; see, for example, [Manski \(2003, 2007, 2013\)](#), [Tamer \(2010\)](#), [Ho and Rosen \(2017\)](#), and [Molinari \(2020\)](#) for reviews.

2. MODEL SET-UP

2.1. Event-study coefficients

We suppose that the researcher has estimated a vector of “event-study coefficients” $\hat{\beta}_n \in \mathbb{R}^{T+\bar{T}}$, which can be partitioned into vectors of coefficients corresponding with the pre-treatment and post-treatment periods, $\hat{\beta}_n = (\hat{\beta}'_{n,pre}, \hat{\beta}'_{n,post})$, where $\hat{\beta}_{n,pre} \in \mathbb{R}^T$ and $\hat{\beta}_{n,post} \in \mathbb{R}^{\bar{T}}$. Event-study estimates of this form arise from non-staggered DiD as well as a variety of related estimators, as we illustrate with several examples

Example 1 (Non-staggered DiD). Consider the canonical DiD setting in which we have a balanced panel of units from period $t = -\bar{T}, \dots, \bar{T}$, and units with $D_i = 1$ receive a treatment beginning in period $t = 1$, while units with $D_i = 0$ never receive the treatment. It is common to report DiD estimates of the form

$$\hat{\beta}_s = (\bar{Y}_{s1} - \bar{Y}_{s0}) - (\bar{Y}_{01} - \bar{Y}_{00}),$$

where $\bar{Y}_{sd}$ is the sample mean of the outcome for units with $D_i = d$ in period $t = s$. Intuitively, $\hat{\beta}_s$ compares the change in the mean outcome between period 0 and period s for the treated and comparison units. In this setting, the estimates $\hat{\beta}_s$ are numerically equivalent to the OLS

coefficients from the regression

$$Y_{it} = \lambda_i + \phi_t + \sum_{s \neq 0} \beta_s \times \mathbb{1}[t = s] \times D_i + \epsilon_{it}. \quad (2)$$

In this case, $\hat{\beta}_{post}$ collects the estimated coefficients corresponding with treated periods, $(\hat{\beta}_1, \dots, \hat{\beta}_{\bar{T}})$, while $\hat{\beta}_{pre}$ collects the estimated coefficients corresponding with periods before treatment $(\hat{\beta}_{-T}, \dots, \hat{\beta}_{-1})$.

Example 2 (Staggered DiD). Event-study coefficients can also be obtained from more complicated DiD procedures. For example, in settings with staggered treatment timing, [Callaway and Sant'Anna \(2020\)](#) propose event-study estimates of the form

$$\hat{\beta}_r = \sum_g w_g \widehat{ATT}(g, g + r),$$

where $\widehat{ATT}(g, t)$ is a DiD estimate that compares the evolution of the outcome for units first treated at period g to units first-treated after period t between time periods $g - 1$ and t , and the w_g are weights that sum to one (*e.g.* proportional to sample size). In this case, $\hat{\beta}_{post}$ collects the values of $\hat{\beta}_r$ for $r \geq 0$ (*i.e.* estimates where one of the groups is treated), and $\hat{\beta}_{pre}$ collects the values of $\hat{\beta}_r$ for values of $r < 0$. Several other related procedures have been proposed for constructing event-study coefficients in contexts with staggered treatment timing; see [de Chaisemartin and D'Haultfœuille \(2021\)](#) and [Roth *et al.* \(2022\)](#) for reviews.

Example 3 (Other related estimators). Other examples of estimators that can be used to produce event-studies coefficients of the form considered here include the GMM procedure proposed by [Freyaldenhoven *et al.* \(2019\)](#), instrumental variables event-studies ([Hudson *et al.* \(2017\)](#)), as well estimators that flexibly control for differences in covariates between treated and comparison groups (*e.g.* [Heckman *et al.* \(1998\)](#); [Abadie \(2005\)](#); [Sant'Anna and Zhao \(2020\)](#)).

2.2. Causal decomposition

Under mild regularity conditions, all of the estimators described above will be asymptotically normally distributed, satisfying $\sqrt{n}(\hat{\beta}_n - \beta) \rightarrow \mathcal{N}(0, \Sigma^*)$ for some parameter vector β . We assume the parameter vector β satisfies the following causal decomposition.

Assumption 1. *The parameter vector β can be decomposed as*

$$\beta = \underbrace{\begin{pmatrix} \tau_{pre} \\ \tau_{post} \end{pmatrix}}_{=: \tau} + \underbrace{\begin{pmatrix} \delta_{pre} \\ \delta_{post} \end{pmatrix}}_{=: \delta} \quad \text{with } \tau_{pre} = 0. \quad (3)$$

The first term, τ , represents the treatment effects of interest. We assume the treatment has no causal effect prior to its implementation, so $\tau_{pre} = 0$. The second term, δ , represents the difference in trends between the treated and comparison groups that would have occurred absent treatment. The parallel trends assumption imposes that $\delta_{post} = 0$, and therefore $\beta_{post} = \tau_{post}$ under parallel trends.³

3. Although our focus is on generalized DiD settings, our results apply to any setting where the researcher has an asymptotically normally distributed estimator $\hat{\beta}$ of a reduced form parameter β that satisfies (3). For example, $\hat{\beta}$ could

2.2.1. Example: non-staggered DiD (continued). Suppose the observed outcome satisfies $Y_{it} = D_i Y_{it}(1) + (1 - D_i) Y_{it}(0)$, where $Y_{it}(1)$ and $Y_{it}(0)$ are, respectively, the potential outcomes when unit i is ultimately treated/not treated. Assume further that there is no anticipation of treatment, so that $Y_{it}(1) = Y_{it}(0)$ for all $t < 1$. Then, for any s , under mild regularity conditions $\hat{\beta}_s$ will be consistent for

$$\beta_s = \tau_{ATT,s} + \underbrace{\mathbb{E}[Y_{is}(0) - Y_{i0}(0) | D_i = 1] - \mathbb{E}[Y_{is}(0) - Y_{i0}(0) | D_i = 0]}_{\text{differential trend} =: \delta_s},$$

where $\tau_{ATT,s} = \mathbb{E}[Y_{is}(1) - Y_{is}(0) | D_i = 1]$ is the average treatment effect on the treated in period s , and δ_s is the difference in trends in potential outcomes between period 0 and period s .⁴ Since the no-anticipation assumption implies that $\tau_{ATT,s} = 0$ for $s < 0$, this yields the decomposition (3).

2.2.2. Example: staggered DiD (continued). Likewise, in the staggered DiD context, define $Y_{it}(g)$ to be the potential outcome for unit i in period t if they are first treated at period g and $Y_{it}(\infty)$ to be the never-treated potential outcome. Then $\hat{\beta}_r$ will be consistent for the parameter $\beta_r = \tau_r + \delta_r$, where $\tau_r = \sum_g w_g ATT(g, g+r)$ and $ATT(g, g+r) = \mathbb{E}[Y_{i,g+r}(g) - Y_{i,g+r}(\infty) | G_i = g]$ is the ATT in period $g+r$ for units first treated at period g . Likewise, $\delta_r = \sum_g w_g \delta_{g,g+r}$, where $\delta_{g,g+r} = \mathbb{E}[Y_{i,g+r}(\infty) - Y_{i,g-1}(\infty) | G_i = g] - \mathbb{E}[Y_{i,g+r}(\infty) - Y_{i,g-1}(\infty) | G_i > g+r]$ is the difference in trends in never-treated potential outcomes between units first treated at period g and units first treated after period $g+r$. Under a no-anticipation assumption, $\tau_r = 0$ for $r < 0$, which again yields the decomposition (3).

2.2.3. Example: other related estimators (continued). We can decompose β as in (3) for other estimators as well. For example, for event-study IVs (with non-staggered timing), τ_{post} is a vector containing the local average treatment effect for each period, and δ represents the asymptotic bias of the IV estimator (*e.g.* from violations of independence or exclusion) for each period. For methods that flexibly control for covariate differences between treated and comparison groups, τ_{post} is again a vector of ATTs, and δ_{post} represents a weighted average (across covariates) of the violation of the conditional parallel trends assumption.

2.3. Target parameter and identification

We suppose the target parameter is a linear combination of the post-treatment causal effects, $\theta := l' \tau_{post}$ for some known $\bar{T}$ -vector l . For example, θ equals the t th period causal effect τ_t when the vector l equals the t th standard basis vector. Similarly, θ equals the average causal effect across all post-treatment periods when $l = (1/\bar{T}, \dots, 1/\bar{T})'$.

We obtain partial identification of θ by assuming that δ lies in a researcher-specified set of possible differences in trends $\Delta \subseteq \mathbb{R}^{T+\bar{T}}$. This nests the usual parallel trends assumption as the special case with $\Delta = \{\delta : \delta_{post} = 0\}$. Since $\delta_{pre} = \beta_{pre}$ is identified, the assumption that

correspond with a vector of first-differences or triple-differences estimators, and δ the violation of the corresponding identifying assumption.

4. We focus on the ATT as the target parameter, as in most of the DiD literature. If one is interested in the population-wide average treatment effect (ATE), one could obtain bounds on the ATE under restrictions on treatment effect heterogeneity, or other assumptions that allow one to bound the treatment effects for untreated units; see [Manski and Pepper \(2013\)](#) for an insightful discussion.

$\delta = (\delta'_{pre}, \delta'_{post})' \in \Delta$ restricts the possible values of δ_{post} given the identified value of the pre-treatment difference in trends δ_{pre} .

It is natural to place restrictions on the relationship between δ_{pre} and δ_{post} , since applied researchers frequently test the null hypothesis $\delta_{pre} = 0$ in order to assess the plausibility of the assumption that $\delta_{post} = 0$. Our identification framework, which generalizes the partial identification framework in [Manski and Pepper \(2018\)](#), thus helps formalize the intuition motivating pre-trends testing.

Under the assumption that $\delta \in \Delta \neq \{\delta : \delta_{post} = 0\}$, the parameter θ will typically be set-identified. The *identified set* is the set of values for θ that are consistent with a given value of β under the restriction $\delta \in \Delta$,

$$\mathcal{S}(\beta, \Delta) := \left\{ \theta : \exists \delta \in \Delta, \tau_{post} \in \mathbb{R}^{\bar{T}} \text{ s.t. } l' \tau_{post} = \theta, \beta = \delta + \begin{pmatrix} 0 \\ \tau_{post} \end{pmatrix} \right\}. \quad (4)$$

When Δ is a closed and convex set, the identified set has a simple characterization.⁵

Lemma 2.1. *If Δ is closed and convex, then $\mathcal{S}(\beta, \Delta)$ is an interval in $\mathbb{R}$, $\mathcal{S}(\beta, \Delta) = [\theta^{lb}(\beta, \Delta), \theta^{ub}(\beta, \Delta)]$, where*

$$\theta^{lb}(\beta, \Delta) := l' \beta_{post} - \underbrace{\left(\max_{\delta} l' \delta_{post}, \text{ s.t. } \delta \in \Delta, \delta_{pre} = \beta_{pre} \right)}_{=: b^{max}(\beta_{pre}, \Delta)}, \quad (5)$$

$$\theta^{ub}(\beta, \Delta) := l' \beta_{post} - \underbrace{\left(\min_{\delta} l' \delta_{post}, \text{ s.t. } \delta \in \Delta, \delta_{pre} = \beta_{pre} \right)}_{=: b^{min}(\beta_{pre}, \Delta)}. \quad (6)$$

Proof. Re-arranging terms in (4), the identified set can be equivalently written as $\mathcal{S}(\beta, \Delta) = \{\theta : \exists \delta \in \Delta \text{ s.t. } \delta_{pre} = \beta_{pre}, \theta = l' \beta_{post} - l' \delta_{post}\}$. The result is then immediate. $\square$

2.3.1. Example: non-staggered DiD (continued). In the three-period DiD model ($\bar{T} = \bar{T} = 1$), the ATT in period 1 is point identified if we assume that the counterfactual post-treatment difference in trends δ_1 is exactly zero (parallel trends). Instead, we assume $\delta = (\delta_{-1}, \delta_1)' \in \Delta$ for some set Δ . When Δ is closed and convex, the identified set for the ATT in period 1 is $[\beta_1 - b^{max}, \beta_1 - b^{min}]$, where $b^{max} = \max_{\delta} \delta_1$ s.t. $(\delta_{-1}, \delta_1)' \in \Delta$ is the maximum possible bias of $\hat{\beta}_1$ given $\delta_{-1} = \beta_{-1}$ and b^{min} is defined analogously.

Additionally, it is immediate from the definition of the identified set in (4) that if Δ is the finite union of sets, $\Delta = \bigcup_{k=1}^K \Delta_k$, then the identified set is the union of the identified sets for its subcomponents,

$$\mathcal{S}(\beta, \Delta) = \bigcup_{k=1}^K \mathcal{S}(\beta, \Delta_k). \quad (7)$$

This fact will be useful, since several empirically relevant choices of Δ can be written as the finite union of convex sets, as we will see below.

5. Our focus is on inference on θ . If one were instead interested in estimating the endpoints of the identified, a natural estimator when Δ is closed and convex would be $[\theta^{lb}(\hat{\beta}, \Delta), \theta^{ub}(\hat{\beta}, \Delta)]$, which is the sample analogue to the bounds derived in Lemma 2.1; see [Supplementary Material, Appendix C](#) for additional discussion.